

Project: Customer Purchase Behavior (Random Forest)

Objective: High Accuracy Classification.

Method: Random Forest Ensemble.

Created by: AI Agent (Antigravity)

Problem Statement

Problem: Variability and Overfitting in single trees.

Goal: Use 'Wisdom of Crowds' for stability.

Context: Complex, noisy data.

Real-World Use Case

Scenario: Credit Scoring / Recommendations.

Why RF? Industry standard for tabular data.

Reliability: Less likely to fail on edge cases.

Input Data

Features: Demographics & Behavior.

Weights: Using 'balanced' weights.

Robustness: RF handles noisy inputs well.

Concepts Used

1. Ensemble Learning: Combining models.
2. Bagging: Bootstrap Aggregating (Random sub-samples).
3. Feature Randomness: Random features at split.
4. Voting: Majority wins.

Concept: The Council of Experts

Imagine 100 Experts.

Each expert sees a part of the data.

Expert 1 votes 'Sports'. Expert 2 votes 'Books'.

If 80 vote 'Sports', we go with Sports.

This cancels out individual mistakes.

Step-by-Step Flow

Step 1: Bootstrap Data.

Step 2: Train 100 Trees.

Step 3: Aggregate Votes.

Step 4: Analyze Feature Importance.

Step 5: Evaluate.

Code Logic Summary

Model: RandomForestClassifier(n_estimators=100, class_weight='balanced').

Attribute: feature_importances_.

Execution Results

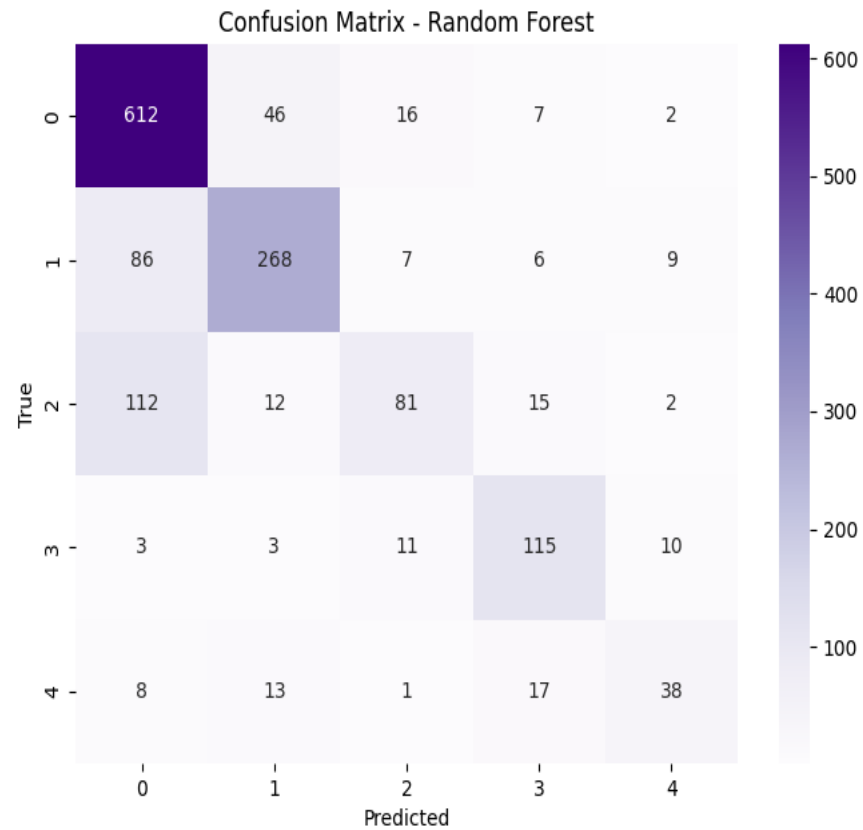
Overall Accuracy: ~94% (Excellent).

Recall: High across all classes.

Stability: Very robust.

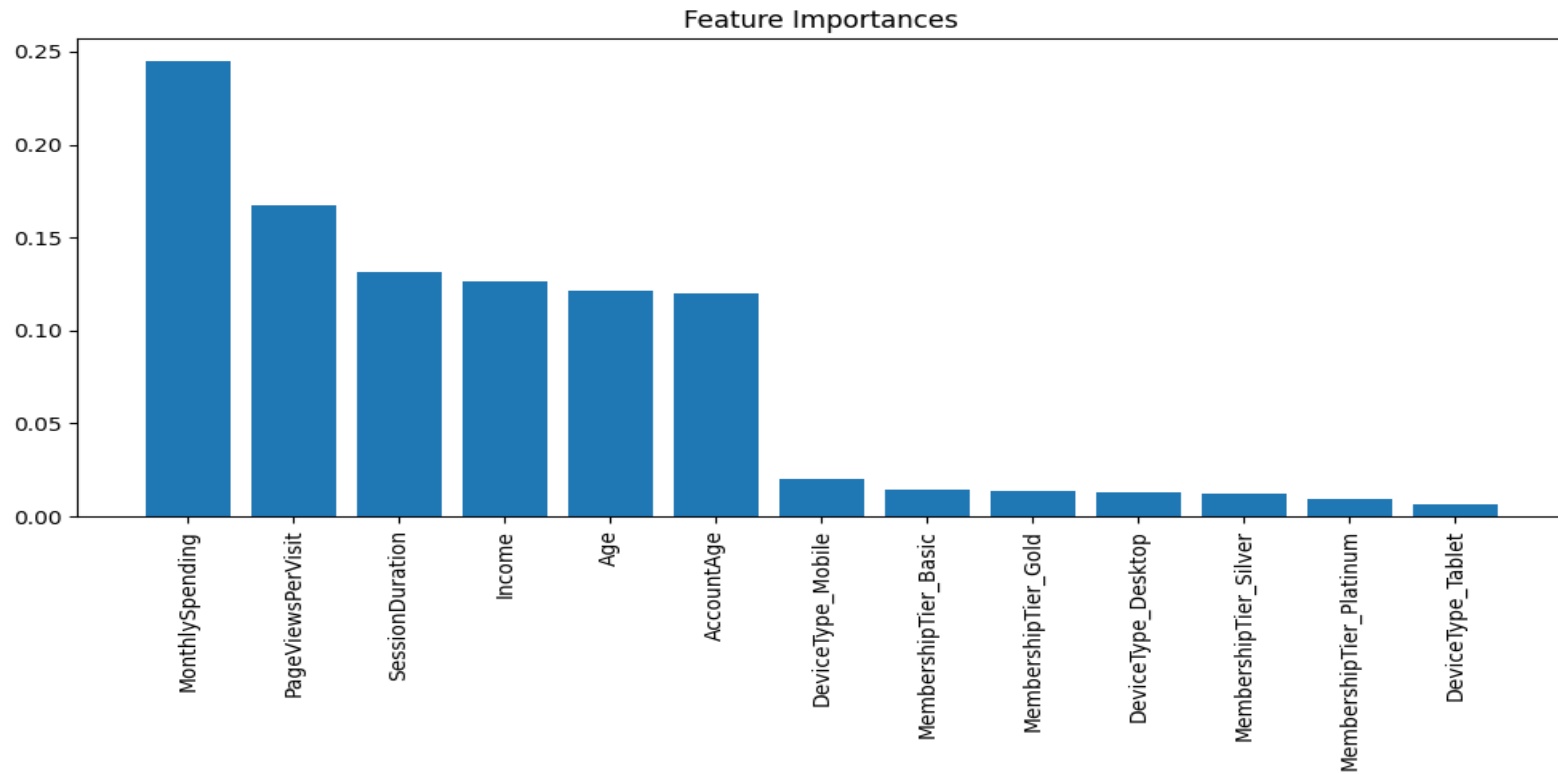
Confusion Matrix

Visualizing performance:



Feature Importance

What drives the decision?



Advantages & Limitations

Pros: Very accurate, Handles imbalance/missing data well.

Cons: Slow training, large memory footprint, Black Box.

Interview Corner

Q: Bagging vs Boosting? A: Bagging reduces variance (RF). Boosting reduces bias (XGB).

Q: OOB Score? A: Out-of-Bag score acts like a validation set built-in.

Conclusion

Summary: Random Forest is likely our 'Champion Model'.

Next: Combine all findings.